Day 1: Introduction to Networking and the OSI Model

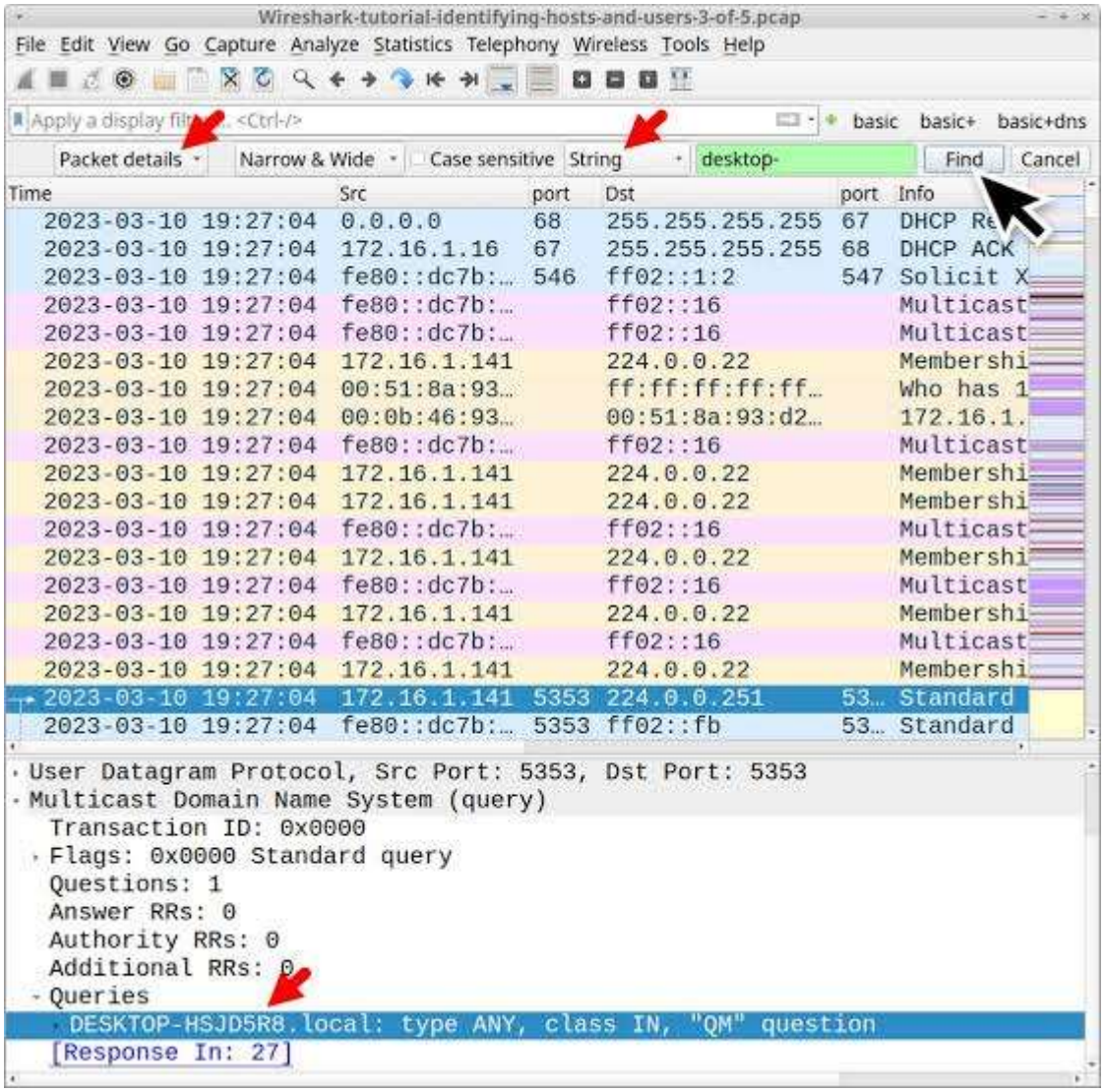
Practical Task Answers

Q1. Install Wireshark on your laptop, start a capture on your main network interface, and visit any website (like instagram.com). Stop the capture and note down the protocol names you see in the packet list.

Steps followed

1. Download Wireshark from the official site wireshark.org and install it on the laptop. Keep the Npcap option ticked during installation, because it is needed to capture packets.
2. Open Wireshark. On the home screen select the main network interface (Wi-Fi or Ethernet), which is the one showing a moving activity graph.
3. Click the blue shark-fin button (Start capturing packets).
4. Open a browser and visit instagram.com. Wait for the page to load.
5. Click the red square button (Stop capturing packets).
6. Look at the Protocol column of the packet list and note down the different protocol names. Useful display filters are dns, tcp and tls.

Screenshot of my capture:



Protocols normally seen in this capture

Protocol	What it does	OSI layer
DNS	Converts the name instagram.com into an IP address.	Application (7)
TCP	Makes a reliable connection (handshake) between my laptop and the server and numbers the data.	Transport (4)
TLSv1.2 / TLSv1.3	Encrypts the data of a secure (HTTPS) website, for example the Client Hello and Application Data packets.	Presentation (6)
QUIC / UDP	Fast transport used by modern sites and apps like Instagram and YouTube instead of TCP.	Transport (4)
HTTP / HTTP2	Carries the web page data (visible only if the site is not encrypted or the packet is decrypted).	Application (7)
ARP	Finds the MAC address of the router from its IP address.	Data Link (2)
ICMP	Sends error and ping messages between devices.	Network (3)
DHCP, mDNS, SSDP	Automatic IP address assignment and discovery of devices (TV, printer) on the local network.	Application (7)

Note: The exact protocols depend on the browser, website and network, so the list above shows the protocols that are normally seen. Write down the protocols from your own capture. Because Instagram uses HTTPS, the page content is encrypted and appears as TLS or QUIC, not plain HTTP.

Q2. Pick any 3 packets from your Wireshark capture and identify which OSI layer each packet mainly represents (for example, HTTP for Application layer, TCP for Transport layer, etc.). Use the Protocol column and Info details in Wireshark to help you map protocols to OSI layers.

Pick any 3 packets from the capture and fill in the Packet No. from your own capture. The table gives typical examples.

No.	Packet No.	Protocol	Info column (typical)	Main OSI layer	How I identified it
1	____	DNS	Standard query A www.instagram.com	Application (7)	DNS is an application protocol. The details pane shows Domain Name System with the name being looked up.
2	____	TCP	52341 to 443 [SYN] Seq=0 Win=64240	Transport (4)	The details pane shows Transmission Control Protocol with source port, destination port 443, sequence number and the SYN flag, which are Transport layer fields.
3	____	TLSv1.3	Client Hello	Presentation (6)	The details pane shows Transport Layer Security. It negotiates encryption, which is the job of the Presentation layer (the data below it is carried by TCP).

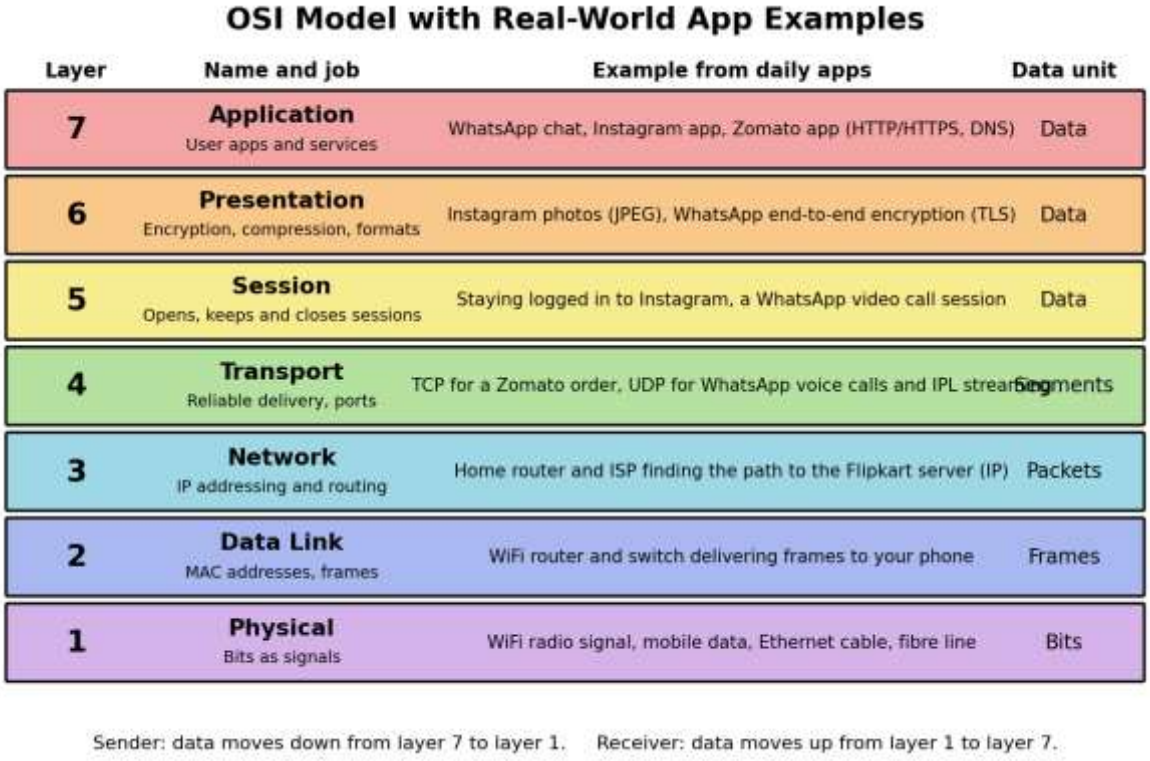
How Wireshark's packet details pane matches the OSI layers

Line in the details pane	OSI layer	What it contains
Frame	Physical (1)	Size of the captured bits and the time

Line in the details pane	OSI layer	What it contains
Ethernet II	Data Link (2)	Source and destination MAC addresses
Internet Protocol Version 4	Network (3)	Source and destination IP addresses, TTL
Transmission Control Protocol / UDP	Transport (4)	Port numbers, sequence number, flags
Transport Layer Security	Presentation (6)	Encryption details of HTTPS
HTTP / DNS	Application (7)	The actual request, such as the website name

Q3 Create a colorful diagram of the OSI Model using any drawing tool (MS Paint, Canva, or even paper and upload a photo). For each layer, add a real-world example from apps you use daily (e.g., WhatsApp uses the Application layer, Instagram images use the Presentation layer, etc.).

The diagram below shows the 7 layers (from the Application layer at the top to the Physical layer at the bottom) with a real-world example from daily apps for each layer. It is made colourful as the task asks.



Layer	Name	Example in daily apps
7	Application	WhatsApp chat window, Instagram app and Zomato app use HTTP/HTTPS and DNS to talk to their servers.
6	Presentation	Instagram photos are compressed as JPEG, and WhatsApp messages are encrypted before they are sent.
5	Session	Staying logged in to Instagram, or a WhatsApp video call, which is one session until it ends.
4	Transport	A Zomato order uses TCP so that nothing is lost; WhatsApp voice calls and IPL streaming use UDP-based transport for speed.

Layer	Name	Example in daily apps
3	Network	IP addresses are used, and the home router and ISP find the route to the Flipkart server.
2	Data Link	The WiFi router and switch use MAC addresses to deliver data to the correct phone or laptop.
1	Physical	WiFi radio signals, mobile data, Ethernet cable and the fibre line carry the bits.

Q4. Explain how sending a message on WhatsApp travels through at least 4 layers of the OSI Model, describing what happens at each layer in your own words. Think about how your message is typed, packaged, sent over the internet, and received.

Example: I type "Are you coming to college today?" in WhatsApp and press Send.

Layer 7 - Application: I type the message in the WhatsApp app and tap Send. The app itself is the application layer: it takes my text and prepares a request for the WhatsApp server.

Layer 6 - Presentation: The message is converted into a standard format and encrypted (end-to-end encryption), so that nobody in the middle can read it. Photos or videos would also be compressed here.

Layer 5 - Session: A session is opened (or kept alive) between my phone and the WhatsApp server so that the two can exchange messages, delivery ticks and the "typing" status.

Layer 4 - Transport: The message is cut into small pieces called segments and each piece is numbered. TCP makes sure that all pieces reach the server in the correct order, and resends any piece that is lost. A port number (443) shows which service should receive it.

Layer 3 - Network: Every piece is wrapped in a packet that has my phone's IP address and the WhatsApp server's IP address. Routers on the internet read the destination IP and choose the best path.

Layer 2 - Data Link: The packet is put in a frame with the MAC address of the next device, which first is my WiFi router. The frame also has a check code for finding errors.

Layer 1 - Physical: The frame is turned into bits and sent as radio waves (WiFi or mobile data), then as electrical or light signals on cables and fibre until it reaches the server.

Receiving: At the server and later at my friend's phone the process is done in reverse. The signals become bits (layer 1), frames (2), packets (3), segments (4), a session (5), the message is decrypted (6), and WhatsApp shows my text in the chat (7). When my friend's phone receives it, the delivery tick appears on my phone.